Do we have enough data? Training counts, market totals and xG

Prepared 6 November 2026. Nothing here changes picks; the new options are in the code, off. Merged to dev.

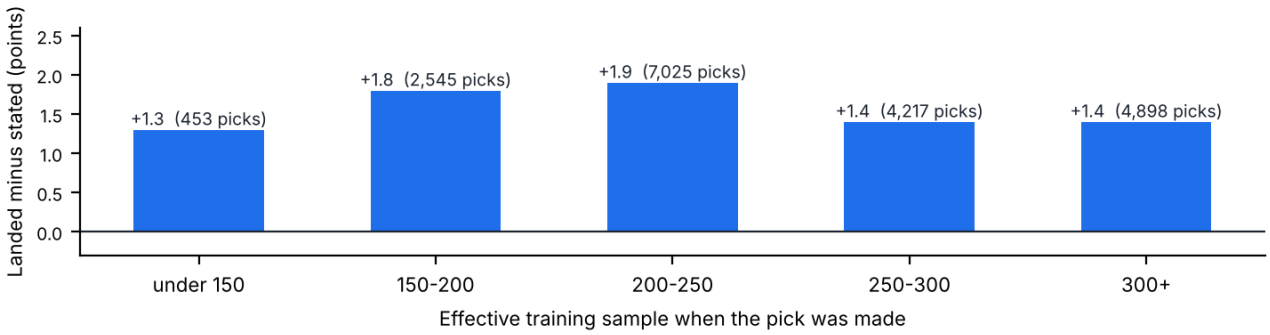
Bottom line. We have enough data for every competition we publish. All 42 football competitions pass the 200-match bar, and a replay of 17,636 picks shows no drop in quality even where the weight-adjusted sample is under 200. Using the market's over/under price improves the model's probabilities clearly but not its picks, so it is not shipped. xG is the best single predictor we have, but we hold it for only eight competitions and it is worth a small gain, so it is added as an off-by-default option.

1. Matches per competition against the 200 bar

The gate counts every stored row however old. The fit forgets with a 180-day half-life, so I also counted the effective sample (each row weighted by its age).

Competition	Rows	Effective
CAF Champions League	1,248	136
Scottish Premiership	1,616	155
Greek Super League	1,731	157
Bundesliga	7,416	196
Ligue 1, NPFL, 2. Bundesliga, Belgian, Primeira, Ligue 2, Eredivisie, Super Lig	2,175 to 8,878	202 to 212
Premier League, Liga MX, Serie B	2,451 to 9,210	241 to 249
Serie A, La Liga, Brasileirao, La Liga 2, League One, Championship	3,365 to 13,416	254 to 372
Argentina, MLS	2,724 and 3,784	386 and 434
Coppa Italia, UEFA cups, English cups (pooled)	1,582 to 4,240	504 to 1,369
National-team competitions (pooled)	13,470 to 25,395	355 to 649

The three thinnest are also the three weakest live leagues (CAF 9 of 23, Greek 65%, Scottish 69% against about 76% stated, 23 to 26 picks each). I checked whether thin data explains it:



It does not. Picks land at or above their stated chance at every effective sample, including under 150. Scottish (effective 159, 738 replay picks) landed at its stated rate. An effective sample of about 150 is enough, so the live shortfalls look like small samples. CAF Champions League has no replay and stays held.

2. Market-informed totals (your point 1)

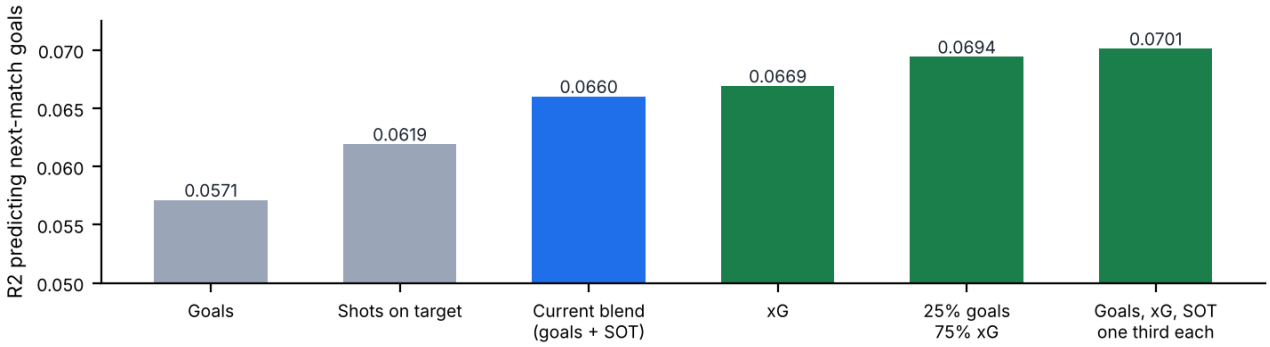
The model's two goal rates are scaled until its over 2.5 equals the closing market's price (margin removed). Tested on 17,857 games, with closing prices standing in for SportyBet's at publish, so this is an upper bound.

Log loss (lower is better)	Current	With market total	Change
Over 1.5 goals	0.5444	0.5377	-0.0067
Over 3.5 goals	0.5936	0.5871	-0.0065
Both teams to score	0.6865	0.6841	-0.0024
Match result (1X2)	1.0081	1.0057	-0.0024
Sep-Oct 2026, over 3.5 goals	0.6445	0.6303	-0.0142

Over 2.5 is set from the price, so it is not a test. The probabilities clearly improve, and on your September and October games most of all, because the market knows goals are up this autumn. **The picks do not improve.** Hit rate with the rules unchanged moved -0.1, -0.1 and -1.3 points on the three samples; with the pick rules refitted on the new numbers it was 77.5% against 77.3% on 2025-26 and 77.6% against 77.8% on Sep-Oct, inside noise. The Code Hub replay got worse (a code landed on 58% of days against 67%) and Forge was mixed. It is the same result as the stretch test: better probabilities, no better picks, because the pick rules are fitted to the old numbers. Shipping it would also mean reading SportyBet's board before predicting, so a game's numbers would depend on whether the board answered in time. Not shipped; the option stays in the code, off.

3. xG (your question)

xG is already stored, for about 7,200 matches in the Premier League, La Liga, Serie A, Ligue 1, Bundesliga and the three UEFA cups since Aug 2023, and for no other competition. I tested how well each measure predicts a team's next-match goals from its last ten (12,584 team-matches):



xG beats goals and shots on target on their own, and the best blend beats the current one by about 6%. For goals conceded xG is also best (0.0256 against 0.0222 for the current blend). By the shots precedent that is worth roughly 0.001 to 0.002 of log loss, on the five leagues where the model is already best. So xG is added as an option (*xgWeight* in the fit, tested, off), but the data is not yet in the training rows: it sits in a different table keyed by match, and three seasons of five leagues is too little to run a decisive walk-forward test. A deeper xG history is the real gate. Given sections 1 and 2, xG's likelier payoff is better explanations on match and team pages than better picks.

What to do

Next step	Why
Leave the 200-match gate as it is	Passes everywhere; no quality loss down to an effective sample of 150.
Keep CAF Champions League held	No replay exists for it, and live it is 9 of 23.
Do not ship the market total for picks	Probabilities improve, picks and codes do not.
Source a deeper xG history before enabling xG	Three seasons of eight competitions cannot confirm a 0.001 to 0.002 gain.

Detail and reproduction: docs/model-audit-teams-2026-11.md, section J (scripts/audit-eval.ts, audit-market-total.ts, decision-lab.ts, forge-lab.ts).